

MOTIVATION

1 Broad Context:

Hydrogen, a clean energy fuel, poses a hidden threat. At just **4%** concentration, it becomes **explosively flammable**. Being invisible and odorless, it can seep through microscopic cracks, turning industrial systems into potential hazards. **The danger is rising—and it's everywhere.**



Select Summaries of Hydrogen Incidents from the H2tools.org Lessons Learned Database 2020
23/05/2019
Hydrogen tanks at a ceramics company at Gangneung Science Park (Solar-to-Hydrogen facility, South Korea) exploded • An engineer was demonstrating the system, feeding hydrogen from the tanks to FC units for generation. • Two people at the site visit were dead, and six were injured.

2 Specific Problems

Hydrogen behaviour



Hydrogen burns with a near-invisible flame under normal conditions

Industrial Requirements

Current hydrogen detection methods are either not effective in locating small leaks in large infrastructures or can not warn us of the leak unless it reaches the saturation point which is **4%**.

Research Background

Hydrogen Detection

Hydrogen is widely used in energy, transportation, and industry. To monitor its presence, researchers have developed sensors based on chemiresistive, electrochemical, and catalytic technologies. These methods are known for their sensitivity and are commonly used in controlled environments.

As hydrogen use expands, especially in pipelines, storage tanks, and vehicles, there's growing interest in materials that are easy to apply over large or irregular surfaces. Flexible, passive, and scalable systems are especially valuable for future hydrogen infrastructure.

Gas Chemochromic Materials

Gas chromic materials respond to gases by changing color due to chemical interaction. When combined with metals like palladium and tungsten oxide, they offer a visual way to detect hydrogen through a color shift—without electronics or power.



Proposeded Mhacansin for change in color
was done by the finalist using Word,2025.

Paint Technology

This paintable sensor offers a practical step forward in hydrogen safety. Its simplicity, flexibility, and visibility make it ideal for future energy systems where traditional sensors may fall short. By combining nanotechnology with everyday usability, it helps bring advanced science closer to real-world solutions.

New Approach

This research presents a paintable Pd-WO₃ nanocomposite that shifts from light gray to blue when exposed to hydrogen. The material can be applied directly to various surfaces, offering a fast, visible, and accessible method to support hydrogen safety in modern systems.

Aim and Novelty

To develop for the first time a gas chromic paintable sensor for hydrogen detection to be painted in a hydrogen surface environment for fast hydrogen gas detection that will enhance the safety system with hydrogen rapid growth as clean energy best candidate

Hypothesis

Using Pd-WO₃ nanoparticles in a flexible, color-changing coating will allow fast, reversible, visible, and accurate detection of hydrogen gas leaks.

Variables

Dependent

.The amount of hydrogen that will be detected at different temperature
.The time it will take to change color upon hydrogen exposure

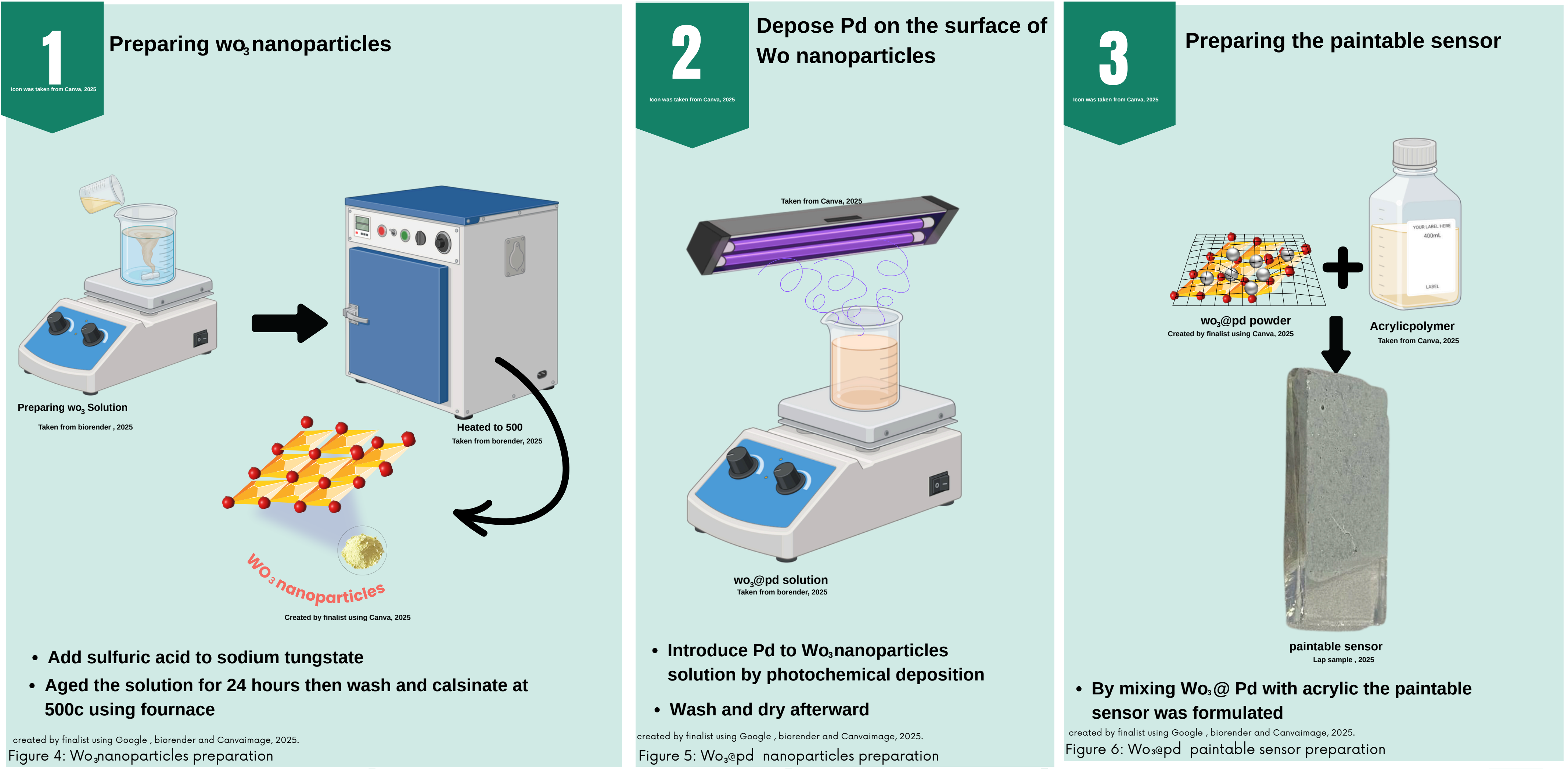
Controlled

Ratio of the WO₃@Pd to the paint remain the same in each paint sample

Independent

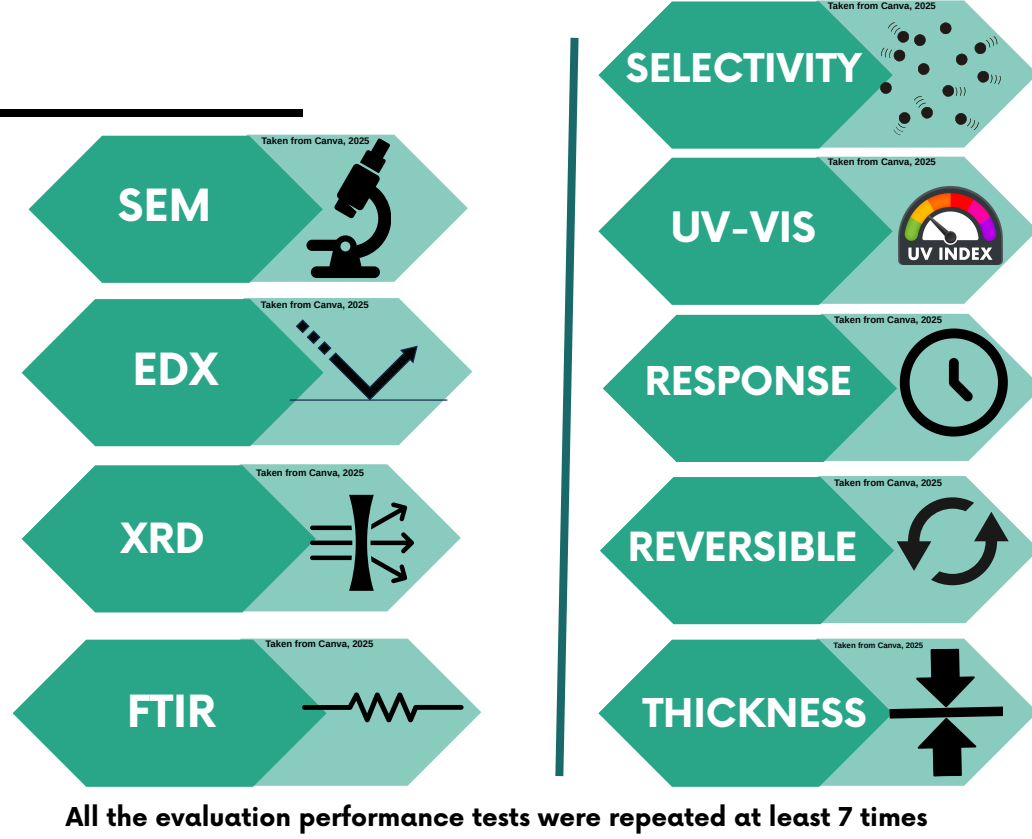
.Temperature (T) during the experiment.
.The pressure and temperature in the hydrogen exposure

Methodology



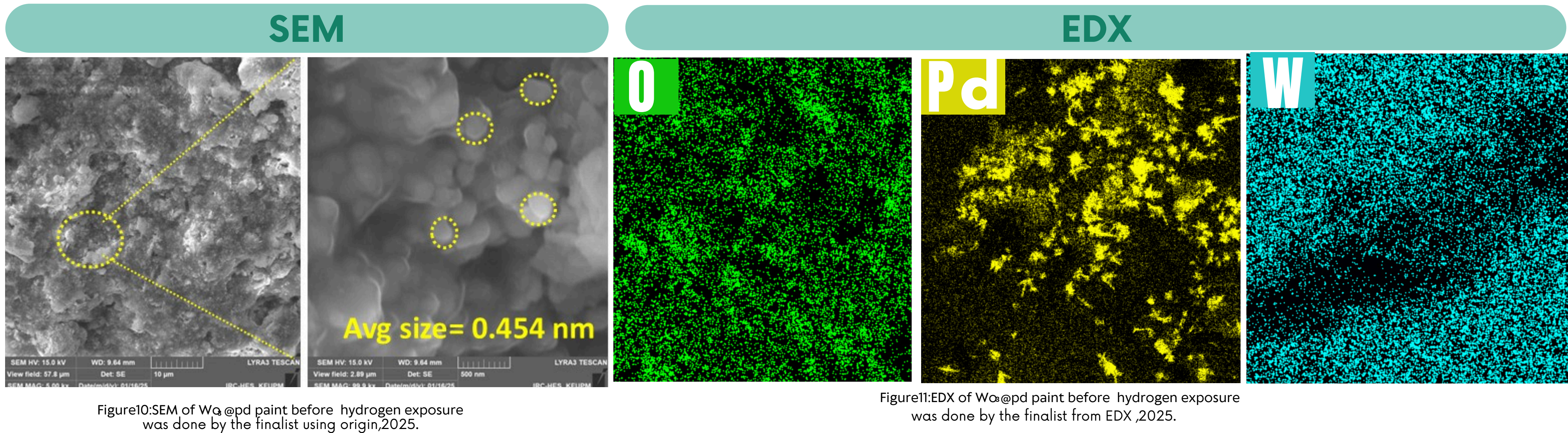
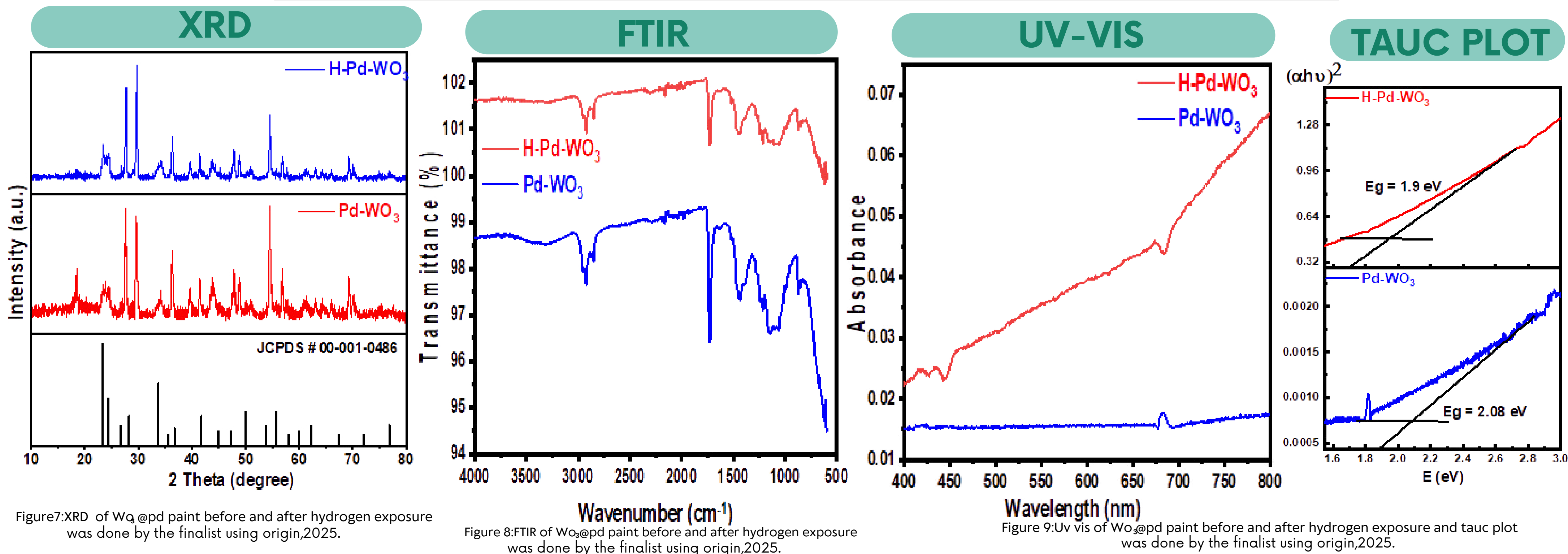
Characterization and Sensor Evaluation

- The sensor nano structure and and response towards hydrogen gas was tested using X-Ray Diffraction (XRD) Fourier Transform infrared spectroscopy (Ftir)(Uv -vis) (Sem)Scanning electron microscopy
- Sensor performance was assessed at a gas station to evaluate sensitivity and selectivity. Time response, reversibility, and color change were tested in a closed reactor. Paint thickness was measured using a coating thickness gauge.

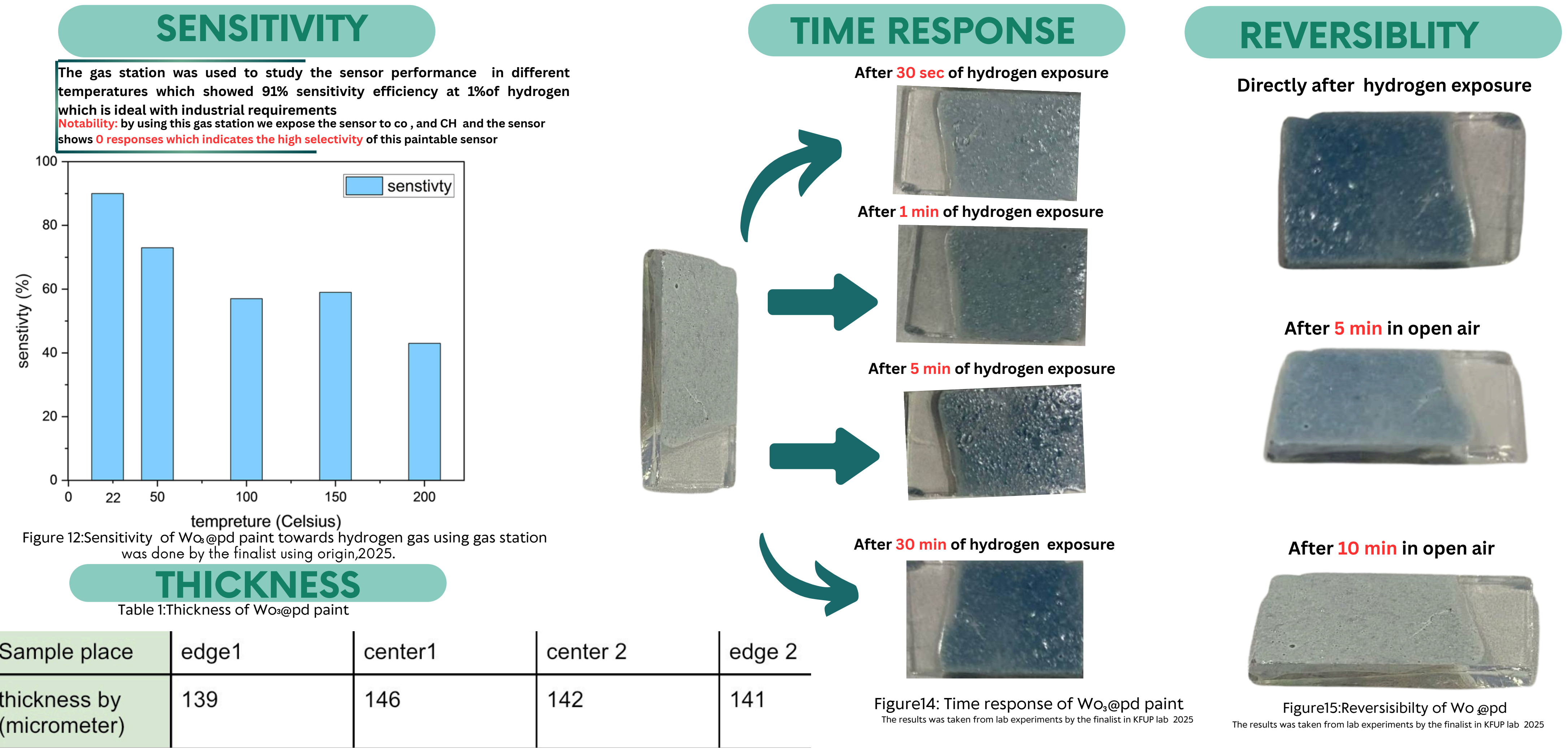


Data & Graphs

Characterization



Sensor Performance



Results

Fig.12 shows High sensitivity, and **selectivity** Achieved 91% **sensitivity** for 1% (less then saturation point)at room temperature demonstrated exceptional selectivity for hydrogen over other gases while fig 15 improves **reversibility** behavior **with** quick reversible response in 10 minutes and fig 14 demonstrate a rapid response time of less than 30 seconds,

SEM EDX Analysis:Fig10 for **SEM** Confirmed the successful synthesis of WO₃@Pd nanoparticles, revealing a uniform nanostructure that enhances the material's surface area and interaction with hydrogen.while fig 11**EDX** proved that we successfully deposited palladium in the surface

Gasochromic Properties: Were validated through XRD (Fig. 7), FTIR (Fig. 8), and UV-Vis (Fig. 9) analyses, which confirmed structural and optical changes after hydrogen exposure.

Comparison and Cost Analysis

Table 2: Cost Analysis for 10×10 cm Paintable Sensor Sample
Created by the finalist using google slide, 2025

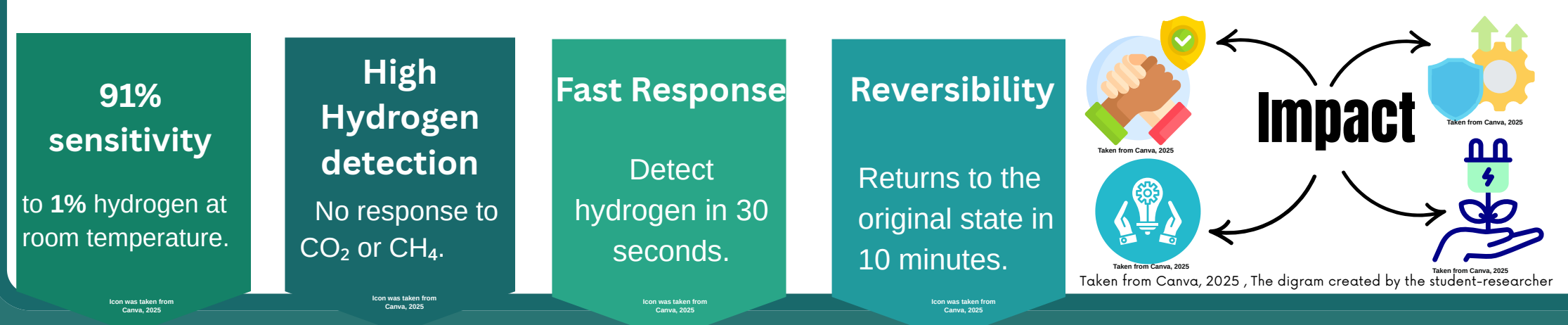
Material	Cost (\$)
Total energy cost	0.0006052
Sodium Tungstate Dihydrate	0.0424875
Sulfuric acid	0.029
DI water	0.000004
Methanol	0.061
Palladium Nitrate	1.79375
polyvinylpyrrolidone	0.002975
polymer resin	0.011
Cost(\$) price of 10*10 cm of the paint	1.9408217

Table3:Compared to different study
Created by the finalist using google slide, 2025

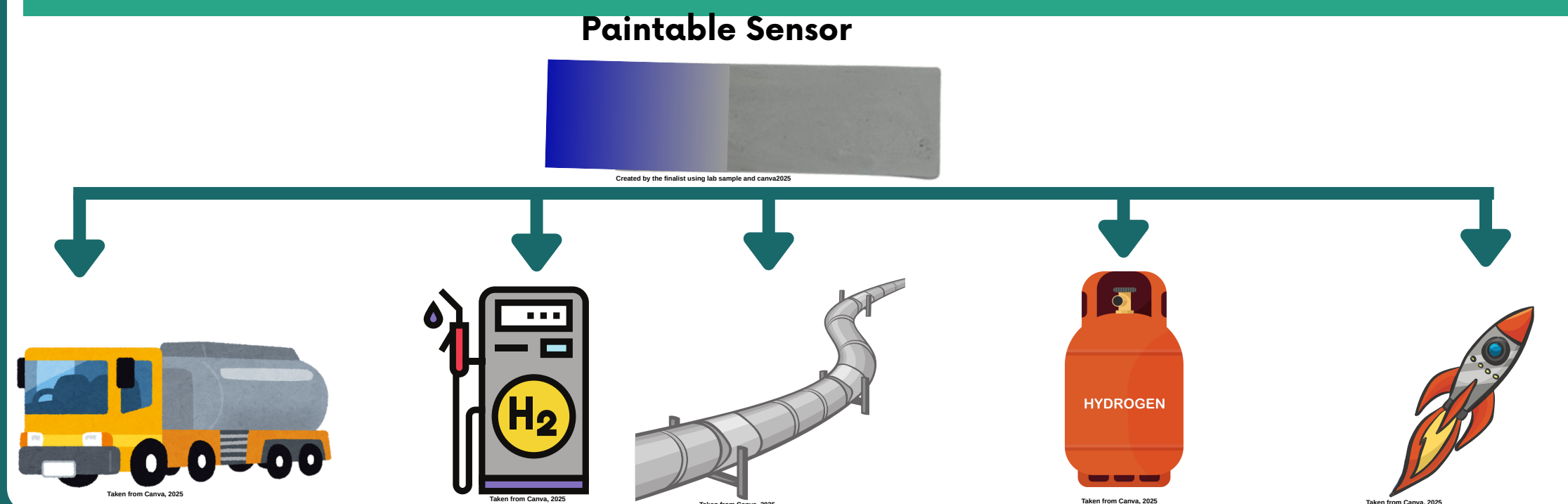
Sensor	application for large infrastructure
Pt-doped TiO ₂ nanotubes	This sensor can detect the hydrogen-based on conductivity it's not an effective approach for large infrastructure it can not locate the hydrogen leakage
wo ₃ @pd paintable sensor(This work)	This sensor is the ideal sensor it can detect the leakage, and the leakage location and cover large areas

Conclusion

The hypothesis has been validated. The developed Pd-WO₃ chemochromic coating provides rapid, color-based hydrogen detection, **improves safety protocols**, validates from safety company and helps prevent losses—potentially **saving over 50 million dollars** from hydrogen-related incidents. Marking a **first step** toward scalable, visual leak detection solutions.



Applications



Future Work

- Connect the innovative sensor with the (MEMs) micro electro-mechanical system for easy monitoring in the control room
- Apply in existing hydrogen tankers and pipelines to enhance hydrogen safety system

References

- Chauhan, P. S., & Bhattacharya, S. (2019). Hydrogen gas sensing methods, materials, and approaches to achieve parts per billion level detection: A review. International Journal of
- Mirzaei, A., Kim, J. H., Kim, H. W., & Kim, S. S. (2019). Gasochromic WO₃ nanostructures for the detection of hydrogen gas: An overview. Applied Sciences (Switzerland), 9, 1775.
- Ndaya, C. C., Javahiraly, N., & Brioude, A. (2019). Recent advances in palladium nanoparticles-based hydrogen sensors for leak detection. Sensors (Switzerland), 19, 4478.
- Yeasmin, R., Jung, G., Han, S., Park, C., & Seo, H. (2024). Self-healing and self-adhesive hydrogen gas sensing tape for robust applications. Chemical Engineering Journal, 482, 148911.

Hydrogen Safety Through Advanced Gas Chromic Coatings for Leak Detection in Next-Generation Storage and Transportation Infrastructure

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